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Affiliated to Anna University, Chennai



Department of Electronics and Communication Engineering

**Course name : Project with Design Thinking /Course code: U23EC651/
Innovative multidisciplinary Project, Project with Design Thinking
Title of the project : Smart Room Occupancy Monitoring System**



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OVERVIEW

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CHALLENGE STATEMENT

In modern smart buildings, classrooms, offices, and large public venues, maintaining safe occupancy levels is essential for ensuring safety, user comfort, and efficient utilization of resources such as lighting, ventilation, and air-conditioning systems. Overcrowding in confined spaces can lead to serious safety hazards, reduced comfort, and unnecessary energy consumption, especially in environments where occupancy directly influences system control.

Most existing occupancy monitoring systems rely on microcontrollers and software-based processing,



Higher Power Consumption



Limited Real-Time Performance



LITERATURE SURVEY

S. No	TITLE	AUTHOR	NAME OF ORGANIZATION & YEAR	DESCRIPTION
1	Infrared-Based Bidirectional Visitor Counter System	S. M. Metev, V. P. Veiko	IEEE / Embedded Systems, 2018	• Uses IR sensors to detect entry and exit based on beam interruption • Implemented using microcontrollers with sequential processing. • Limited accuracy during fast movement.
2	Arduino-Based Smart Room Occupancy System	A. Kumar, R. Singh	International Journal of Engineering Research, 2019	• Low-cost system using Arduino and IR sensors • Easy implementation and suitable for small applications. • Suffers from processing delays and timing issues.
3	IoT-Based Occupancy Monitoring and Energy Control System	M. S. Hossain, G. Muhammad	IEEE IoT Conference, 2020	• Integrates occupancy detection with IoT for remote monitoring. • Controls lighting and ventilation automatically. • Depends on network connectivity and increases system complexity.
4	8051 Microcontroller-Based Visitor Counter	R. S. Gaonkar	Embedded Systems Journal, 2017	• Uses basic IR sensors with 8051 controller • Simple design for classrooms and offices. • Not suitable for real-time high-speed movement detection.
5	Computer Vision-Based People Counting System	Z. Zhang, Y. Wang	IEEE Transactions / Computer Vision, 2021	• Uses cameras and deep learning for high accuracy. • Works well in crowded environments • High power consumption, cost, and privacy concerns.
6	Smart Building Occupancy Detection System	J. A. Stankovic, I. Lee	Smart Cities Journal, 2022	• Focuses on energy-efficient building management. • Uses multiple sensors for occupancy tracking • Requires complex integration and higher system cost.

S. No	TITLE	AUTHOR	NAME OF ORGANIZATION & YEAR	DESCRIPTION
7	Technique Bidirectional Visitor Counter	S. R. Kumbhar, P. M. Patil	ResearchGate, 2023	• Uses IR sensors to detect entry and exit based on beam interruption • Implements sequence logic with FSM • Low-cost and simple design
8	FPGA Implementation of People Counter Using Algorithm	A. Manduchi, et al.	University of California, Santa Cruz, 2023	• Uses image processing with FPGA for real-time counting • Applies background subtraction and tracking • High-speed parallel processing
9	Bi-Directional Visitor Counter using IoT	R. Sharma, K. Verma	ACM Digital Library, 2024	• Integrates IR sensors with IoT modules (ESP8266/ESP32) • Cloud-based monitoring system • Suitable for smart buildings
10	FPGA Counter Design and Optimization	M. Patel, S. Desai	IJARMT Journal, 2025	• Focuses on efficient counter design using FPGA • Uses synchronous sequential circuits • Optimized for speed and reduced delay
11	Real-Time People Counting System Using FPGA and IoT	K. R. Prakash, L. N. Reddy	ResearchGate, 2023	• Combines FPGA with IoT for real-time monitoring • Uses FSM and up/down counter logic • Supports cloud dashboard integration
12	AI-Based Smart Occupancy Detection System	J. Wang, Y. Liu	IEEE / Elsevier (Recent Trends), 2026	• Uses AI and computer vision for accurate detection • Works in crowded environments • Higher computational cost and complexity

EXISTING METHOD

- Most existing occupancy monitoring systems are based on microcontrollers such as Arduino, ESP32, etc.
- Infrared (IR) sensors are commonly used to detect entry and exit by sensing beam at doorways.
- The system determines direction by analyzing the sequence of sensor activation using software logic.
- All processing is performed using sequential execution, where instructions are executed one after another.
- These systems rely on interrupt-based handling to detect sensor triggers and update the count.
- The occupancy count is updated in software variables, which may introduce delays during execution.
- Some systems are integrated with IoT modules to enable remote monitoring and control.
- Advanced solutions use camera-based systems with computer vision and deep learning for improved accuracy.
- These systems often require continuous power and processing, increasing overall energy consumption.
- The overall system performance depends heavily on software efficiency and processor speed, making it less reliable for real-time and high-speed movement scenarios.

PROPOSED / INNOVATIVE METHOD

Hardware-Centric Design Approach:

- The replace software-driven processing with a fully hardware-based architecture, improving speed and reliability.

Modular RTL Architecture:

- The design is divided into reusable hardware modules, enabling easy expansion and customization.

Fault Isolation Capability:

- Failure in one unit does not affect the operation of other rooms, improving system reliability.

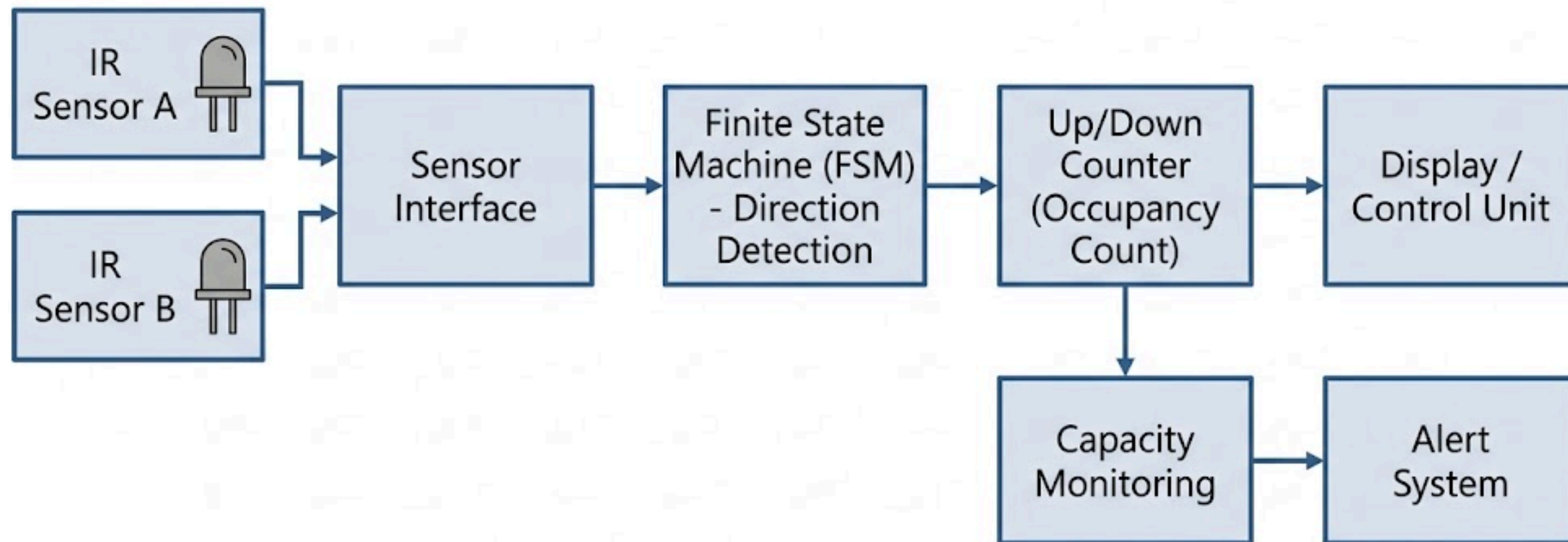
ASIC-Ready Implementation:

- Designed at RTL level, enabling smooth transition from FPGA prototype to ASIC fabrication.

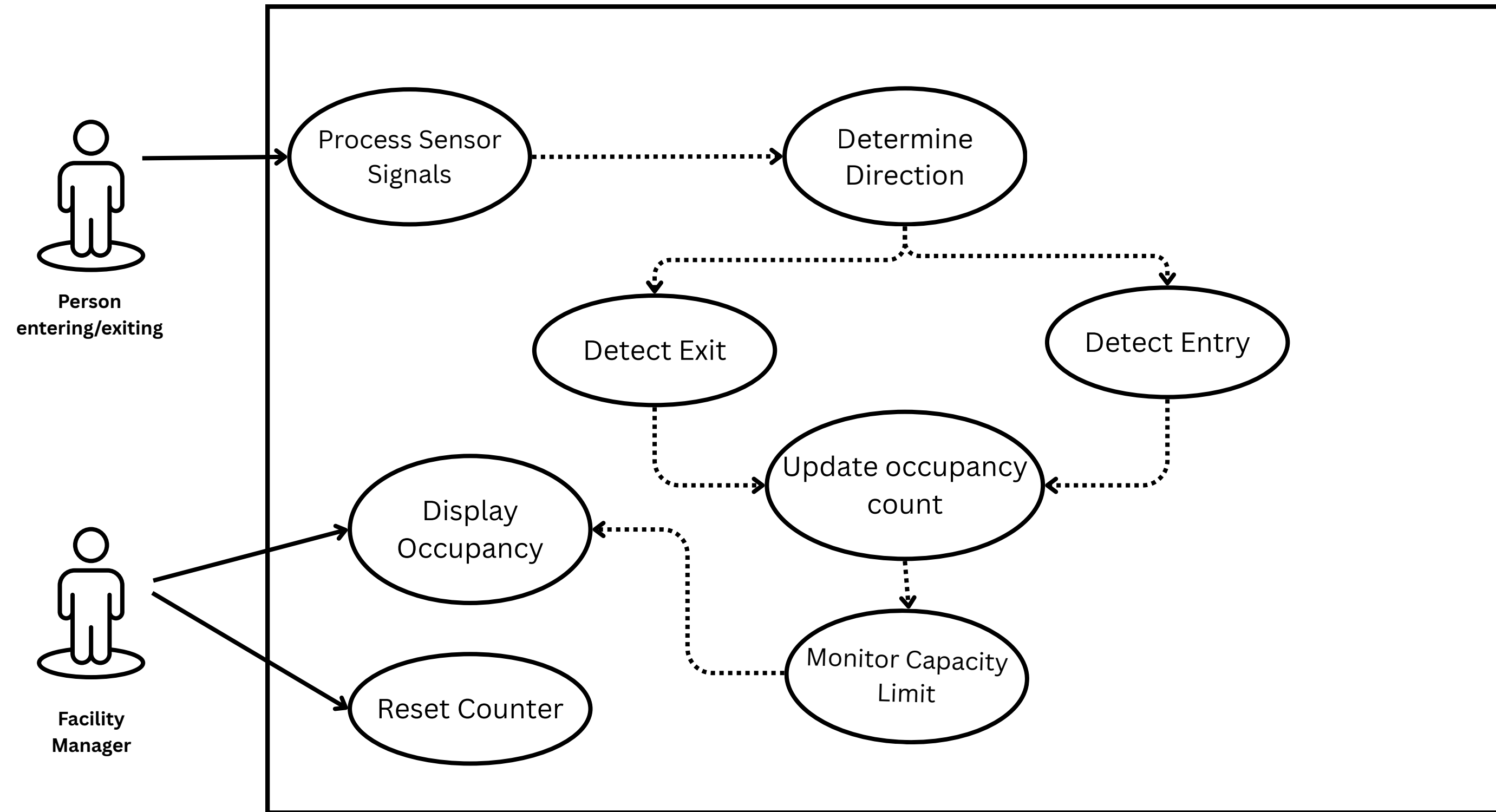
Simplified System Integration: (optional)

- Can be easily integrated with building automation systems for lighting and ventilation control.

BLOCK / SCHEMATIC DIAGRAM



USE CASE DIAGRAM

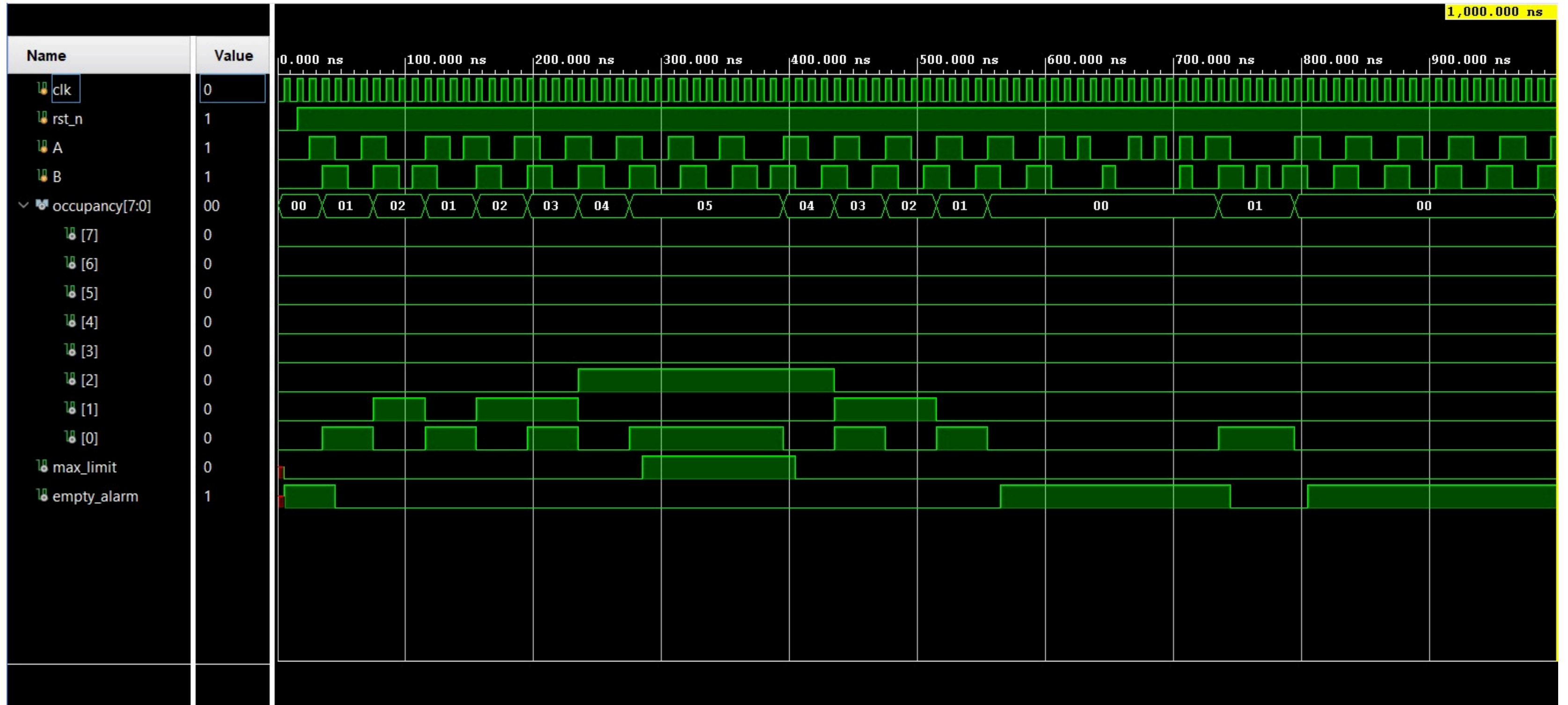


DESCRIPTION OF SOFTWARE/HARDWARE IMPLEMENTATION

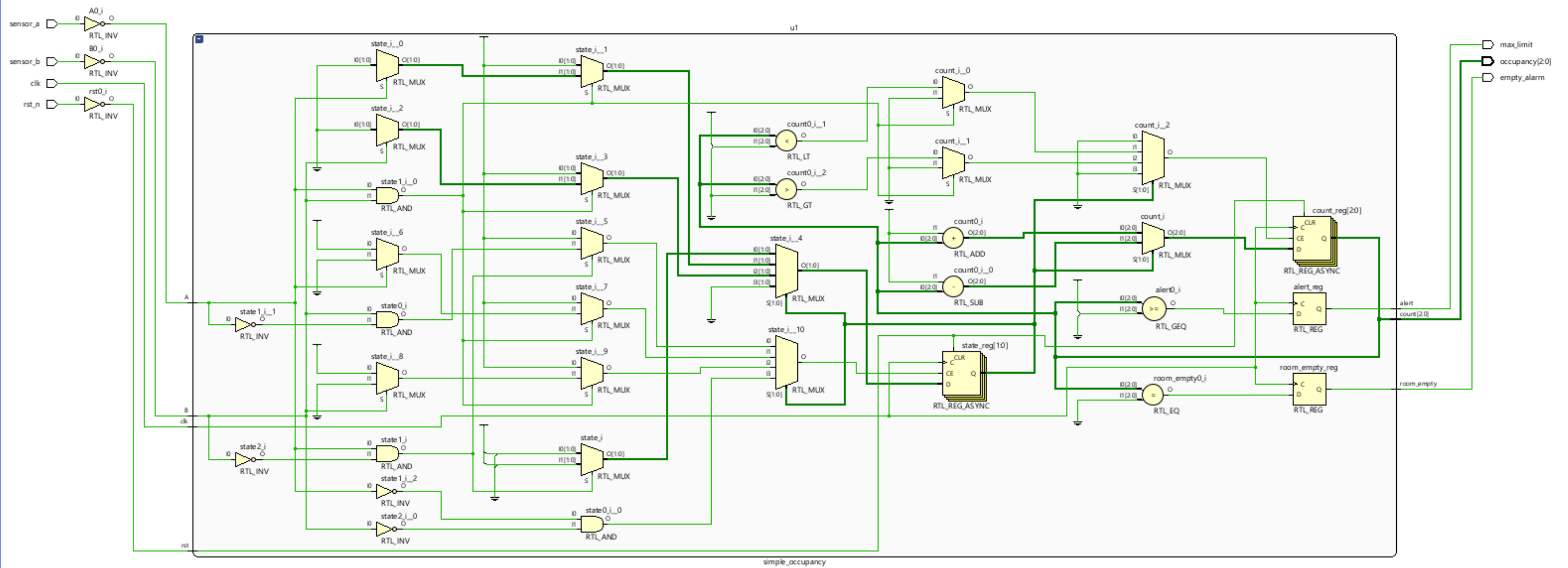
- Start
- Initialize System
 - Set count = 0
 - Set MAX_LIMIT = 5
 - Configure IR sensors A and B as inputs
 - Configure LEDs as outputs
 - Configure reset toggle switch
- Check Reset
 - If reset button is pressed:
 - Set count = 0
 - Turn OFF red LED
 - Turn ON empty indication (if used)
 - Go to Step 3
- Read Sensor Inputs
 - Continuously monitor IR sensors A and B
- Detect Entry (A → B)
 - If sensor A is triggered first
 - Then sensor B is triggered:
 - If count < MAX_LIMIT
 - Increment count = count + 1
- Detect Exit (B → A)
 - If sensor B is triggered first
 - Then sensor A is triggered:
 - If count > 0
 - Decrement count = count - 1
- Invalid Conditions Handling
 - If only A or only B is triggered → Ignore
 - If A and B are triggered simultaneously → Ignore
 - If incomplete sequence → Ignore
- Update Output
 - Display count in binary on LEDs
 - If count == 0
 - Indicate room is empty (optional LED)
 - If count >= MAX_LIMIT
 - Turn ON red LED (room full)
 - Else
 - Turn OFF red LED
- Wait for Sensors to Reset
 - Ensure both sensors return to LOW
 - Prevent double counting
- Repeat Steps 3–9 Continuously
- Stop

EXPERIMENTAL RESULTS - SIMULATION

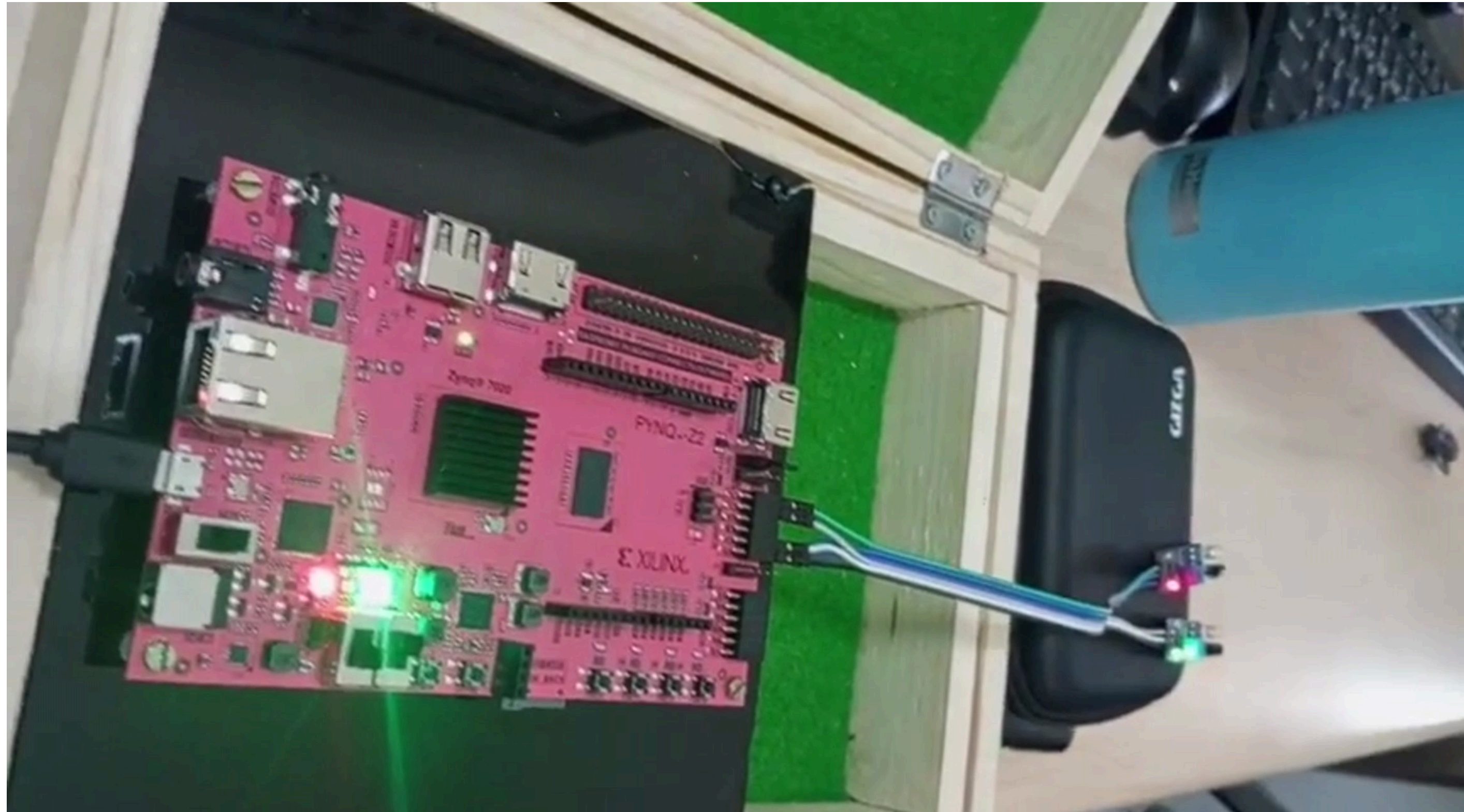
WAVEFORM



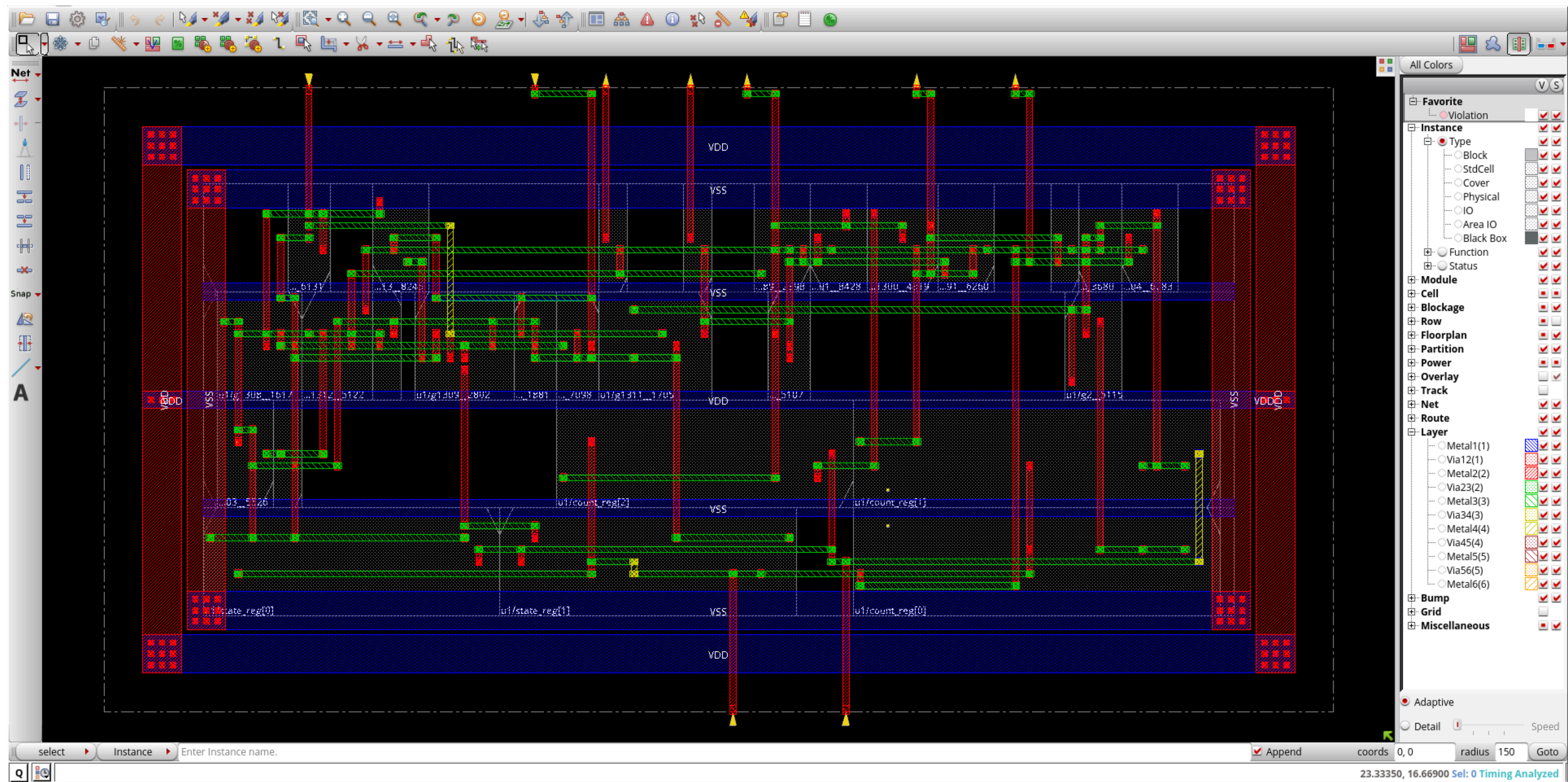
COMPLETE HARDWARE DESIGN



SOFTWARE/HARDWARE IMPLEMENTATION VIDEO



ASIC IMPLEMENTATION



Power Consumption Measurements

1. The basic empty `setup() {} loop() {}` program

ASIC POWER CONSUMPTION

Here are my results:

	Uno	Nano	Mega
Revision	R3	R3	R3
CPU	ATmega328	ATmega328	ATmega2560
Voltage	5.10V	5.10V	5.10V
Program #1 Current	144mA	35mA	79mA
Program #1 Power	734mW	179mW	403mW

Instance: /top
Power Unit: W
PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	1.07204e-08	3.63425e-05	4.51019e-06	4.08634e-05	74.74%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	7.97556e-09	4.60378e-06	5.78704e-06	1.03988e-05	19.02%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	3.41172e-06	3.41172e-06	6.24%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	1.86960e-08	4.09463e-05	1.37089e-05	5.46739e-05	100.00%
Percentage	0.03%	74.89%	25.07%	100.00%	100.00%

From tlextrait.svbtle.com

CONCLUSION

- The proposed system provides an efficient and reliable solution for real-time room occupancy monitoring.
- It uses a hardware-based design (Verilog HDL) instead of software-based approaches.
- FPGA validation ensures correct real-time operation before ASIC implementation.
- Overcomes limitations of existing systems such as: Sequential delays, Timing uncertainties
- Reduced accuracy during continuous movement
- The Finite State Machine (FSM) enables accurate bidirectional entry and exit detection.
- Hardware-level processing ensures deterministic and low-latency performance.
- Supports capacity monitoring and energy management for smart building applications.
- The modular architecture improves flexibility and ease of design.
- The one chip per room approach enhances scalability, reliability, and fault isolation.
- The design is ASIC-oriented, making it suitable for large-scale deployment.
- Provides a low-power and privacy-preserving solution (no camera-based tracking).

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